

LDL & CARDIAC ARREST

Introduction

A worldwide study demonstrated that among all modifiable risk factors of CV disease, abnormal serum levels of cholesterol were associated with the highest attributable risk for the occurrence of CVD, especially ischemic heart disease (Eujene Jung: et al, 2022)

A lipoprotein is a biochemical assembly whose primary function is to transport fat molecules in water, as in blood plasma or other extracellular fluids (Eujene Jung: et al, 2022)

LDL-C has been identified as the main risk factor for CVD by many epidemiological and interventional studies because LDL-C plays a major role in the pathogenesis of atherosclerosis (Eujene Jung: et al, 2022)

Recently, LDL-C has generally replaced TC as the primary lipid measurement for predicting cardiovascular risk (Eujene Jung: et al, 2022)

An abundance of evidence has demonstrated the causal role of low-density lipoprotein cholesterol (LDL-C) in the development and progression of atherosclerotic cardiovascular disease (ASCVD), and therefore, measuring LDL-C level plays a central role in risk assessment in all international guidelines. some patients may display resilience to the detrimental effects of LDL-C. (Martin Bodtker Mortensen: et al, 2023)

Long-term high LDL levels have been linked to heart attacks, other cardiac disorders, and the development of blood vessel plaque. (Manya Singh, 2023)

Circulating LDL particles are able to penetrate the endothelium of arterial walls and become oxidized, promote inflammation, and consequently develop severe atherosclerosis. (Rishi K. Wadhera: et al, 2016)

A high LDL level is associated with an increased risk for heart disease and stroke. LDL transports cholesterol to the arteries, and when the LDL level is elevated, this cholesterol can accumulate in the blood vessel walls and contribute to the formation of a plaque. This plaque formation, or atherosclerosis, can eventually cause the vessel to narrow and decrease blood flow to the heart (coronary artery disease). If the vessel becomes blocked so that blood no longer reaches the heart or brain, this can result in chest pain (angina), heart attack, or stroke. (Sally Robertson, 2020)

In rare cases, people have a very high LDL level as the result of an inheritable genetic disease called hypercholesterolemia that reduces the liver's ability to clear excess cholesterol. The very high LDL level can lead to heart attack and stroke among younger individuals (Sally Robertson, 2020).

atherosclerotic cardiovascular disease (ASCVD) as its primary component, is the leading cause of death in the USA. low-density lipoprotein cholesterol (LDL-C) in particular (the so-called 'bad' cholesterol), is a root cause of ASCVD. Despite the generally accepted belief that 'lower LDL-C is better,' meta-analyses indicate that high LDL-C is associated with at most a small increased absolute risk of ASCVD or premature mortality. (Kevin E kip: et al, 2024).

Cardiovascular disease (CVD) remains the leading cause of death worldwide. Elevated low-density lipoprotein cholesterol (LDL-C) levels are clearly associated with increased risks of CVD death and incident CVD events in multiple cohort studies. (Shuang Rong: et al, 2022).

It remains controversial whether reductions in cardiovascular mortality after intensive lowering of low-density lipoprotein cholesterol (LDL-C) depend on baseline LDL-C levels. To reassess these findings, in this brief report, we performed an updated literature search through February 2020 and selected randomized controlled trials which reported cardiovascular mortality and major adverse cardiovascular events (MACE) as outcomes. (Safi U Khan: et al, 2021).

Sudden cardiac arrest (SCA) is defined as unexpected sudden loss of cardiac contraction that occurs without a warning sign, majority of SCA occurs in patients with structural heart disease, including coronary artery disease, heart failure and cardiomyopathy. Besides established structural heart disease, we have identified several independent risk factors of SCA from previous studies, which includes hypertension, diabetes mellitus, dyslipidemia, and metabolic syndrome. Co-existence of diabetes with dyslipidemia elevates the risk of cardiovascular disease, which is also known as diabetic dyslipidemia. Although diabetes mellitus is also known as an independent predictor of SCA from previous studies, there have been conflicting results on association of dyslipidemia with SCA risk. (Yun Gi Kim: et al, 2023)

Methods

(Cholesterol or HDL-cholesterol or LDL-cholesterol) and (heart arrest or cardiovascular death or cardiovascular mortality or cardiac death or cardiac mortality) (Eujene Jung: et al, 2022) – search strategy and selection criteria.

The exposure was total serum cholesterol, HDL-cholesterol, and LDL-cholesterol levels measured before the cardiovascular event, and the outcome was cardiovascular disease (CVD) mortality, including coronary heart disease (CHD) mortality. We excluded studies if they did not meet these criteria. (Eujene Jung: et al, 2022) – study selection.

We used data from the Western Denmark Heart Registry (WDHR) to evaluate the association of LDL-C with myocardial infarction (MI) and ASCVD events according to presence versus absence of coronary atherosclerosis. Data for this study were provided by WDHR. Our study included all patients >18 years of age who underwent CTA between January 1, 2008, and December 31, 2017, in Western Denmark for the indication of symptoms suggestive of CAD.

Exclusion criteria were known CAD (previous MI or coronary revascularization) or a missing LDL-C or CTA measurement. Before CTA, all patients underwent a non-contrast-enhanced ECG-gated CAC scan. We defined the presence of coronary atherosclerosis as $CAC > 0$ and absence of coronary atherosclerosis as $CAC = 0$. In sensitivity analyses, we defined the absence of coronary atherosclerosis as $CAC = 0$ combined with no noncalcified plaque, and the presence of coronary atherosclerosis as any detectable atherosclerosis on CTA ($CAC > 0$ or presence of noncalcified plaque). The 2 outcomes used for this study were MI and a composite ASCVD outcome consisting of fatal and nonfatal MI or ischemic stroke. The study outcomes were identified through linkage among national registries covering all Danish hospitals. The association between LDL-C and the presence of CAC was assessed using logistic regression. The association between LDL-C and MI and ASCVD was examined using Cox proportional hazards regression adjusted for age, sex, diabetes, hypertension, CCI, smoking, and pre-CTA statin use. The association between LDL-C and MI and ASCVD was examined on a continuous scale using restricted cubic splines incorporated in Cox proportional hazards models. These analyses were performed in the overall population and stratified by presence versus absence of CAC. The association of LDL-C with ASCVD events (MI, coronary death, resuscitated cardiac arrest, revascularization, or stroke) was evaluated in MESA. (Martin Bodtker Mortensen: et al, 2023)

We conducted a retrospective cohort study of adults aged 50–89 years with hospital and/or office visit data captured through the University of Pittsburgh Medical Center (UPMC) electronic medical record (EMR) system. The date period for analysis was 4 January 2000 through 31 December 2022. Adults without diabetes aged 50–89 years not on statin therapy at baseline or within 1 year and classified as primary prevention-type patients. To mitigate potential reverse causation, patients who died within 1 year or had baseline total cholesterol (T-C) ≤ 120 mg/dL or LDL-C < 30 mg/dL were excluded. Baseline LDL-C categories of 30–79, 80–99, 100–129, 130–159, 160–189 or ≥ 190 mg/dL. All-cause mortality with follow-up starting 365 days after baseline cholesterol measurement. (Kevin E kip: et al, 2024).

This prospective cohort study included a nationally representative sample of 14 035 adults aged 18 years or older, who participated in the National Health and Nutrition Examination Survey III 1988–1994. LDL-C levels were divided into 6 categories: < 70 , 70–99.9, 100–129.9, 130–159.9, 160–189.9 and ≥ 190 mg/dL. Deaths and underlying causes of deaths were ascertained by linkage to death records through December 31, 2015. Weighted Cox proportional hazards regression models were used to estimate the hazard ratios (HR) of mortality outcomes and its 95% CIs. (Shuang Rong: et al, 2022).

We included 53 randomized controlled trials (329,897 patients) of LDL-C lowering therapies (statin, ezetimibe and PCSK9 inhibitors) and stratified the meta-analysis according to the baseline LDL-C thresholds. (Safi U Khan: et al, 2021).

The K-NHIS is the single, exclusive medical insurance system managed by Korean government, that mandates subscription for Korean citizens, including virtually entire Korean population. A regular, nationwide health medical examination. patients with diagnostic codes for type 2 diabetes mellitus and those who underwent medical examination during 2009 to 2012 were included in this study, followed up from the day of initial medical examination to December, 2018. Sudden cardiac arrest was identified with ICD-10 codes; I46.0 (cardiac arrest with successful resuscitation), I46.1 (sudden cardiac arrest), I46.9 (cardiac arrest, cause unspecified), I49.0 (ventricular fibrillation and flutter), R96.0 (instantaneous death), and R96.1 (death occurring less than 24 h from onset of symptoms). Out-of-hospital cardiac arrest declared at emergency department was defined as SCA event, and events during in-hospital admission were not counted. LDL-cholesterol (mg/dL) was calculated using the Friedewald formula. LDL-cholesterol level was classified into eight quantiles: LDL-cholesterol < 70 (group I), $70 \leq$ LDL-cholesterol < 75 (group II), $75 \leq$ LDL-cholesterol < 100 (group III), $100 \leq$ LDL-cholesterol < 115 (group IV), $115 \leq$ LDL-cholesterol < 130 (group V), $130 \leq$ LDL-cholesterol < 145 (group VI), $145 \leq$ LDL-cholesterol < 160 (group VII), $160 \leq$ LDL-cholesterol (group VIII). Cox-proportional hazards model was used to calculate hazards ratios and 95% confidence intervals (CI). Variables that were statistically different according to LDL-cholesterol level were further included for adjusted Cox regression analysis. Consequently, adjustment with covariates was done with multivariate model 1 (age and sex) and model 2 (age, sex, income, body mass index [BMI], smoking status, alcohol consumption status, regular exercise, hypertension, fasting blood glucose, duration of diabetes mellitus, use of insulin, use of oral hypoglycemic agent, and use of statin). (Yun Gi Kim: et al, 2023).

Results

Baseline characteristics of the 23 132 patients included in this study. Overall, 12 340 patients (53%) had CAC=0 and 10 792 (47%) had CAC>0. Over a median follow-up of 4.3 years (interquartile range, 2.4 to 6.1), 275 patients (1.2%) experienced a first MI, and 552 (2.4%) experienced a first ASCVD event (271 MIs and 281 strokes). The numbers of MI and ASCVD events were 35 and 94 in patients with CAC=0 compared with 96 and 192 events in those with CAC>0. A very high LDL-C level (≥ 193 mg/dL versus <116 mg/dL) increased the odds of having CAC>0 versus CAC=0 [likelihood of having atherosclerosis]. For a very high LDL-C level (≥ 193 mg/dL versus <116 mg/dL), the multivariable adjusted HR was 2.34 (95% CI, 1.51–3.62) for MI events and 1.54 (95% CI, 1.10–2.15) for ASCVD events. There was a significant interaction effect of CAC and LDL-C level on MI and ASCVD events. The higher the LDL-C level, the higher the risk for MI and ASCVD in patients with CAC>0 but not in those with CAC=0. In the 12 340 patients with CAC=0, the multivariable adjusted HR per 38.7 mg/dL higher LDL-C level was 1.13 (95% CI, 0.90–1.40) for MI events and 1.02 (95% CI, 0.87–1.18) for ASCVD events. [not significantly associated with an increased risk of events.]. Patients with very high LDL-C level (≥ 193 mg/dL versus <116 mg/dL) was 3.53 (2.11–5.92) among patients

with CAC>0 [The increased risk associated with very high LDL-C was also much more pronounced in patients with CAC>0.]. The incidence event rate in patients with an LDL-C level ≥ 193 mg/dL versus <116 mg/dL was 12.36 (95% CI, 8.06–19.00) in patients with CAC>0 [The incidence rates of MI and ASCVD were much higher in patients with CAC>0, especially in those with higher LDL-C levels.]. All traditional risk factors were associated with events in the overall population and in patients with CAC>0. In those with CAC=0, age, diabetes, current smoking, and HDL-C level were significantly associated with ASCVD events. In sensitivity analyses, we defined absence of coronary atherosclerosis as no detectable atherosclerosis on CTA (CAC=0 including no noncalcified plaque). In total, 10 707 patients had no CTA-defined atherosclerosis, whereas 12 425 patients had detectable atherosclerosis (CAC>0 or presence of noncalcified plaque). In the 3340 individuals with CAC>0, the multivariable adjusted HR per 38.7 mg/dL higher LDL-C was 1.15 (P=0.004) compared with 0.91 (P=0.26) for the 3361 individuals with CAC=0 (Pinteraction=0.02 in fully adjusted model) [the findings are not specific to the Danish population and may be generalizable]. (Martin Bodtker Mortensen: et al, 2023)

177 860 patients with a mean (SD) age of 61.1 (8.8) years and mean (SD) LDL-C of 119 (31) mg/dL were evaluated over a mean of 6.1 years of follow-up. A U-shaped relationship was observed between the six LDL-C categories and mortality with crude 10-year mortality rates of 19.8%, 14.7%, 11.7%, 10.7%, 10.1% and 14.0%, respectively. Adjusted mortality HRs as compared with the referent group of LDL-C 80–99 mg/dL were: 30–79 mg/dL (HR 1.23, 95% CI 1.17 to 1.30), 100–129 mg/dL (0.87, 0.83–0.91), 130–159 mg/dL (0.88, 0.84–0.93), 160–189 mg/dL (0.91, 0.84–0.98) and ≥ 190 mg/dL (1.19, 1.06–1.34), respectively. Unlike LDL-C, both T-C/HDL cholesterol (high-density lipoprotein cholesterol) and triglycerides/HDL cholesterol ratios were independently associated with long-term mortality. (Kevin E kip: et al, 2024).

During 304 025 person-years of follow up (median follow-up 23.2 years), 4458 deaths occurred including 1243 deaths from CVD. At baseline, mean age was 41.5 years and 51.9% were women. Very low and very high levels of LDL-C were associated with increased mortality. After adjustment for age, sex, race and ethnicity, education, socioeconomic status, lifestyle factors, C-reactive protein, body mass index, and other cardiovascular risk factors, individuals with LDL-C<70 mg/dL, compared to those with LDL-C 100–129.9 mg/dL, had HRs of 1.45 (95% CI, 1.10–1.93) for all-cause mortality, 1.60 (95% CI, 1.01–2.54) for CVD mortality, and 4.04 (95% CI, 1.83–8.89) for stroke-specific mortality, but no increased risk of coronary heart disease mortality. Compared with those with LDL-C 100–129.9 mg/dL, individuals with LDL-C ≥ 190 mg/dL had HRs of 1.49 (95% CI, 1.09–2.02) for CVD mortality, and 1.63 (95% CI, 1.12–2.39) for coronary heart disease mortality, but no increased risk of stroke mortality. (Shuang Rong: et al, 2022).

Our meta-analysis found that each 38.7 mg/dL (1 mmol/L) lowering in LDL-C reduced the risk of cardiovascular mortality (RR, 0.85; 95% CI, 0.81–0.89), but this varied by baseline LDL-C of those in the trials (P = 0.04 for interaction). The risk reduction in cardiovascular mortality was

limited to trials with baseline LDL-C of >100 mg/dL. In contrast, the reduction in MACE was independent of baseline LDL-C levels. These findings were consistent in primary and secondary prevention settings for both outcomes and by sex for MACE. Our results support the professional cholesterol guidelines which recommend achieving a $\geq 50\%$ reduction in LDL-C from baseline for high-risk patients. (Safi U Khan: et al, 2021).

A total of 2,602,577 patients were included, with total follow-up duration of 17,851,797 person * year. Mean follow-up duration was 6.86 years, and 26,341 SCA cases were identified. Overall incidence of SCA was highest in the lowest LDL-cholesterol group (< 70 mg/dL) and decreased in a linear manner as LDL-cholesterol rises, till 160 mg/dL. Adjustment of covariates resulted in U-shape association, with highest risk of SCA in the highest LDL-cholesterol group (≥ 160 mg/dL) followed by lowest LDL-cholesterol group (< 70 mg/dL). In subgroup analysis, U-shape association between SCA risk and LDL-cholesterol was more pronounced in male, non-obese people, and those who did not use statins. (Yun Gi Kim: et al, 2023).

Discussion

This collection of studies shows a complex picture of the relationship between LDL-C and cardiovascular outcomes, challenging the simplistic "lower is better" paradigm. While elevated LDL-C is clearly implicated in the development and progression of atherosclerosis and cardiovascular events, including MI, ASCVD, and cardiovascular death, the research presented here consistently demonstrates a more nuanced relationship, often characterized by a U-shaped association between LDL-C and risk. This means that both high and, importantly, *low* LDL-C levels are associated with increased risk of adverse outcomes, including all-cause mortality and sudden cardiac arrest (SCA).

The findings of Mortensen et al. (2023) highlight the importance of considering the presence of coronary artery calcium (CAC) when evaluating LDL-C's impact. Their observation that elevated LDL-C was associated with increased risk of MI and ASCVD *only* in the presence of CAC suggests that LDL-C's detrimental effects might be contingent on existing subclinical atherosclerosis. This has significant implications for risk stratification and could refine targeted interventions for LDL-C lowering.

The studies by Kip et al. (2024) and Rong et al. (2022), both demonstrating a U-shaped relationship between LDL-C and mortality, challenge the prevailing focus solely on LDL-C reduction. These findings suggest that excessively low LDL-C levels may also be detrimental, raising questions about the mechanisms underlying this increased risk. It underscores the need for a more comprehensive understanding of the biological roles of LDL-C and the potential consequences of drastically lowering its levels.

The meta-analysis by Khan et al. (2021) adds another layer of complexity. While they confirmed the overall benefit of LDL-C lowering in reducing cardiovascular mortality, they found this benefit to be primarily observed in trials with baseline LDL-C > 100 mg/dL. This raises the

question of whether the same degree of benefit applies to individuals with lower baseline LDL-C. Interestingly, the reduction in MACE was independent of baseline LDL-C, suggesting a potentially broader benefit for MACE reduction through LDL-C lowering therapies.

Finally, Kim et al. (2023)'s study on SCA in a diabetic population reinforces the importance of considering LDL-C in the context of SCA risk. Their finding of a U-shaped association, with increased SCA risk at both high and low LDL-C levels, highlights the need for careful LDL-C management in this high-risk group. The co-existence of diabetes and dyslipidemia, often referred to as diabetic dyslipidemia, is a well-established risk factor for cardiovascular disease, and these findings suggest that LDL-C management plays a critical role in mitigating SCA risk in this population. The conflicting results on the association between dyslipidemia and SCA risk, as noted by Kim et al. (2023), further emphasize the need for research in this area.

Conclusion

These studies collectively demonstrate that the relationship between LDL-C and cardiovascular outcomes is far more intricate than a simple linear association. The consistent observation of U-shaped curves suggests that maintaining an optimal LDL-C range, rather than simply targeting the lowest possible level, is crucial for cardiovascular health. The findings highlight the importance of personalized lipid management strategies that consider individual patient characteristics, such as the presence of CAC, diabetes status, and the specific clinical context. Further research is needed to elucidate the mechanisms underlying the increased risk associated with low LDL-C levels, to refine LDL-C thresholds for intervention, and to develop more tailored approaches to lipid management that optimize cardiovascular outcomes while minimizing potential adverse effects. Specifically, further investigation into the conflicting results regarding the association between dyslipidemia and SCA risk, as well as the interplay between diabetes, LDL-C, and SCA, is warranted. This research should inform clinical guidelines and lead to more effective strategies for preventing cardiovascular events, including SCA.

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